

EKNOOR KAUR SANDHU

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Computational biology | Biomaterials | Agricultural AI | Community-centered leadership

EDUCATION

SGGS Collegiate Public School, Chandigarh, India | CBSE; expected graduation May 2027

- Transferred after Grade 10 to combine Biology, Mathematics, Artificial Intelligence, and Economics — the school's only student with this combination, and a path outside NEET and JEE, India's national entrance examinations for medicine and engineering.

RESEARCH (COMPUTATION, BIOLOGY, COMPUTER VISION, BIOINFORMATICS)

Crop-Row Gap Detection from Drone Imagery | Paid Research Intern, DronaMaps with Plaksha University | Grade 12

- Flew drones over fragmented smallholdings, recorded ground truth, and detected row breaks and missing plants in RGB imagery covering about 1,790 acres. Co-author; presented at the International Conference on Agriculture-Centric Computation (ICA 2026), Ropar.

Sugarcane Maturity from Satellite Data | Student Researcher, Plaksha University | Grade 11

- Asked whether satellite imagery could help sugar mills schedule harvests more fairly; dropped an image-based maturity claim after an agronomist showed that sucrose accumulates inside the stalk. Presented at Punjab Agricultural University.

Silver Nanoparticle-Silk Fibroin Hydrogels | Research Intern, Plaksha University | Grade 12

- Synthesized cocoon-derived silk fibroin hydrogels loaded with silver nanoparticles; tested swelling, silver release, and antibacterial activity. Found 50 µg/mL comparable to 25 mg/mL, a dose 500 times higher, and reported the result although it contradicted the hypothesis.
- First author; manuscript submitted to the *Journal of Emerging Investigators* (under review).

Bioinformatics | Self-directed and mentored learning | Grades 11–12

- Completed online coursework in Bioinformatics (UC San Diego) and Algorithmic Genomics (Johns Hopkins); arranged access to the Bioinformatics Department, Sri Guru Gobind Singh College, for BLAST searches and alignments; taught a sequence-analysis workshop to 30 classmates.

Independent Coding | Self-directed | Grades 9–12

- Built pdf-repeat-remover (Python, perceptual hashing), dance-face-detect (TensorFlow computer vision), po-di-ng (multiplayer Pong in Odin with TCP networking, built without AI tools), and fast-todo (REST API with FastAPI and SQLite).

LEADERSHIP AND COMMUNITY IMPACT

Head Girl, Student Council | SGGS Collegiate Public School | Grade 12 (Secretary, Grade 11)

- Nominated by students across all four senior streams; presented management evidence of staffing and timetable gaps, after which the school hired four teachers (two AI, one Economics, one counsellor).
- Delegated sponsor relationships to the incoming Secretary; that student-led team raised about USD 7,600 for MUN and the Annual Function.

President, Environment Club / Lead, Green Campus Programme | SGGS Collegiate Public School; APSWDP climate education network | Grade 12

- Replaced awareness drives with student audit teams tracking classroom electricity, waste segregation, and single-use plastic; model adopted through APSWDP across 29 Chandigarh schools.
- Selected by APSWDP as youth representative to the UN Climate Change Conference (COP31), Antalya, November 2026.

President, Dance Club | SGGS Collegiate Public School

- Trained 17 dancers for school functions and 2 state-level competitions (won twice in folk, once in Western); organized 5 environmental street plays.

WORK AND SERVICE EXPERIENCE

Volunteer, Intern, and Master Trainer | Association of Professional Social Workers and Development Practitioners (APSWDP) | Grade 9–present

- Supported supervised, confidential outreach documentation and follow-ups for vulnerable communities in Chandigarh.
- Documented follow-ups on clinic referrals, HIV/STI testing, identity documents, child education, and welfare schemes; helped facilitate 997 ABHA health IDs.

Entrepreneurship Intern | Punjab Innovation Mission | Grade 11

- Supported 6 startup founders across agri-food, edtech, health, and logistics with market research, competitor mapping, and prototyping; presented to 240 attendees.

Annotation Quality Intern | Labellerr AI | Grade 11

- Built Python audit scripts for COCO, Pascal VOC, JSON, and CSV exports, checking 477 files for missing labels, path errors, and empty samples.

HONORS AND SCHOLARSHIPS

AP Scholar with Distinction | College Board | 2026

- Prepared independently for five AP examinations, with no AP instruction offered at school.

STEM Scholar | Junior Academy, New York Academy of Sciences | Global Innovation Challenge | Grade 12

- Designing an energy-retrofit solution with a mentored international team.

Cultural Talent Search Scholarship (Kathak) | Ministry of Culture, Government of India | Grades 9–12

- Seven years of Kathak training; Prachin Kala Kendra, First Division.

Young Technology Scholar | Plaksha University | Grade 9

- Selected from about 700 applicants across 437 schools for a two-week program in AI, design thinking, and research framing.

SKILLS AND INTERESTS

- **Technical:** Python, TensorFlow, Java, BLAST. **Languages:** English, Punjabi, Hindi, French. **Art:** sketching, watercolour, stone painting.